

PATIENT CASE-TAKING SOFTWARE

using AI (Mangaluru Hospitals)

Problem Statement ID: 26047

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1. Introduction

- A web-based patient case-taking platform designed for hospitals in Mangaluru.
- Captures structured patient information, symptoms, evidence and appointment preferences.
- Uses an AI assistant to analyse entered details and ask relevant follow-up questions.
- Connects patients, doctors and hospital authorities through role-based workflows.
- Supports multilingual interaction and voice assistance in Tulu, Kannada, English and Hindi.

2. Problem Statement

Current Challenges

- Manual case-taking can be time-consuming.
- Patient information may be incomplete or inconsistent.
- Doctor availability and appointment allocation need coordination.
- Language barriers can affect communication.

Proposed Need

- Structured digital case records.
- AI-assisted questioning before consultation.
- Capacity-aware doctor assignment.
- Patient-selected appointment slots.
- Secure role-based access.

3. Objectives

- Digitize the patient case-taking process from registration to appointment.
- Improve completeness of patient information using AI-generated follow-up questions.
- Allow patients to select suitable check-up date and time.
- Assign another doctor when the selected/assigned doctor reaches capacity or becomes unavailable.
- Provide configurable doctor recommendations based on case information.
- Support patient, doctor and hospital-authority login and workflows.
- Provide verification-code based patient authentication and account management.
- Enable multilingual voice interaction for Tulu, Kannada, English and Hindi.

4. Existing Systems and Proposed Approach

Existing Approach

- Paper/manual case forms.
- Separate appointment coordination.
- Limited structured patient history.
- Language and accessibility challenges.
- Doctor allocation often handled manually.

Proposed System

- Centralized digital case record.
- AI-assisted case analysis.
- Capacity-aware doctor allocation.
- Integrated appointment scheduling.
- Multilingual voice assistance.
- Role-based hospital portal.

5. Requirements Analysis

Functional Requirements

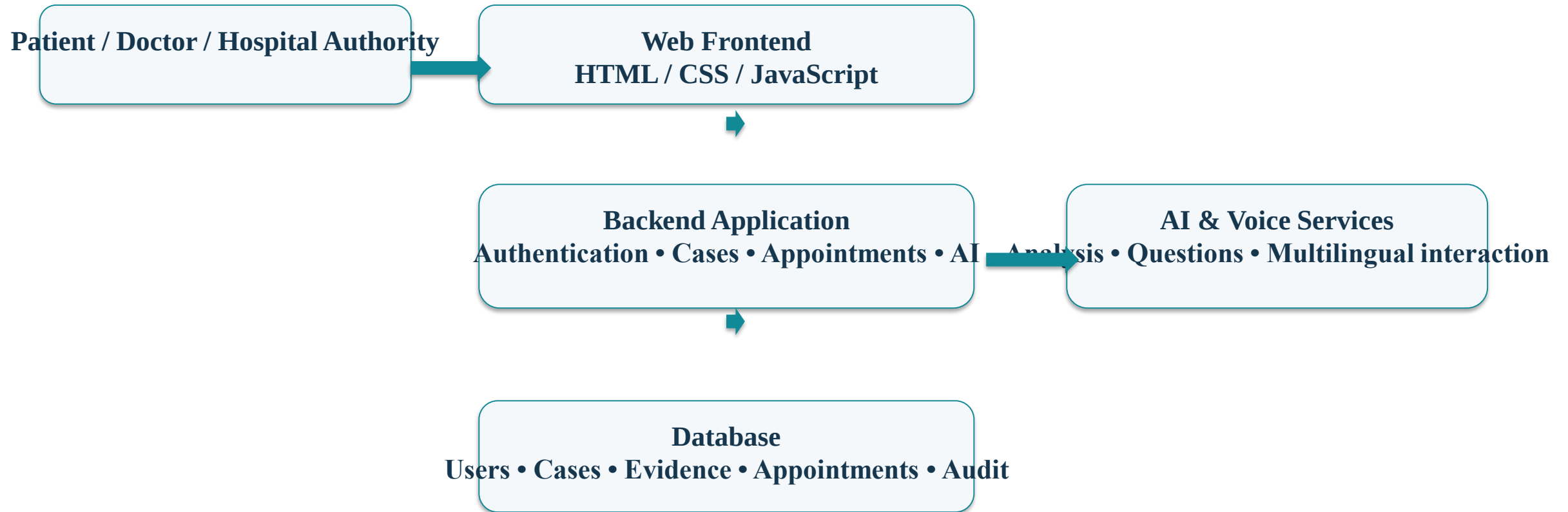
- Patient registration and verification-code login.
- Patient profile and case creation.
- Symptom and evidence capture.
- AI assistant and follow-up questions.
- Doctor recommendations and reassignment.
- Appointment booking and time selection.
- Doctor and hospital-authority portals.

Non-Functional Requirements

- Security and access control.
- Usability and responsive interface.
- Reliable data persistence.
- Auditability of important actions.
- Scalability for multiple hospitals/doctors.
- Maintainable modular architecture.

6. System Architecture

Architecture



7. Core Modules

- Patient Module — registration, verification code, profile, case creation, evidence and appointments.
- AI Assistant Module — analyses case details, identifies missing information and asks follow-up questions.
- Doctor Module — views assigned cases, reviews patient information and manages consultation availability.
- Hospital Authority Module — manages hospital/doctor availability, appointments and operational records.
- Recommendation Module — presents suitable doctor options based on case and availability.
- Appointment Module — supports patient-selected date/time and capacity-aware allocation.
- Audit & Evidence Module — records important actions and stores supporting case material.

8. AI-Assisted Case Taking

AI Workflow

- Read structured patient details.
- Review symptoms and available evidence.
- Identify information gaps.
- Ask contextual follow-up questions.
- Organize responses into the case record.
- Support doctor review before consultation.

Important Design Principle

- AI assists documentation and case intake.
- It does not replace clinical diagnosis.
- Final medical decisions remain with qualified doctors.
- Emergency cases should be directed to emergency care.

9. Doctor Allocation & Recommendations

- Each doctor can have a configurable workload/capacity.
- When the selected doctor is full or unavailable, the system checks alternative available doctors.
- The patient can be offered another suitable doctor rather than leaving the appointment unassigned.
- Doctor recommendations can be configured according to the hospital workflow.
- Availability is considered together with the patient's selected appointment time.
- The allocation process is recorded for traceability.

10. Appointment & Multilingual Voice Support

Patient Experience

Appointment Flow

- Patient completes case details.
- Patient selects preferred date.
- Patient selects available time slot.
- System checks doctor availability/capacity.
- Doctor is assigned or an alternative is offered.
- Appointment confirmation is stored.

Voice / Language

- English
- Kannada
- Tulu
- Hindi
- Voice interaction can assist patients during case entry.
- Language support improves accessibility for local users.

11. User Roles & Security

Access Control

Patient

- Verification-code login.
- Create and manage cases.
- Use AI assistant.
- Book appointments.
- Manage account and request deletion.

Doctor / Hospital Authority

- Doctor: view assigned cases and availability.
- Hospital authority: manage doctors, appointments and hospital operations.
- Role-based authorization limits access.
- Audit records support accountability.

12. Database Design

- Users — identity, role, contact information and authentication-related data.
- Hospitals — hospital identity and Mangaluru-area information.
- Cases — patient case details, symptoms and case status.
- Evidence — supporting files/records associated with a case.
- Appointments — patient, doctor, case, date/time and booking status.
- Leaves / Availability — doctor availability and scheduling constraints.
- Audit — timestamped actions for important system events.

13. Technology Stack

Implementation

Application Layer

- Frontend: HTML, CSS and JavaScript.
- Backend: Python-based web application.
- Database: relational database schema.
- REST-style application endpoints.
- Browser-based responsive interface.

Supporting Services

- AI-assisted analysis and question generation.
- Voice/language interaction layer.
- Secure authentication and session handling.
- Evidence/file handling.
- Audit logging and deployment tooling.

14. Testing Strategy

- Unit testing for authentication, case handling, doctor allocation and appointment logic.
- Integration testing across patient, doctor and hospital-authority workflows.
- Validation of verification-code login and role-based access.
- Appointment tests for available, full and unavailable doctors.
- AI workflow tests for missing-information questions and case updates.
- Multilingual/voice interaction checks.
- Usability and responsive-interface testing.

15. Future Scope

- Integration with hospital information systems and electronic medical records.
- More advanced speech-to-text and text-to-speech support for regional languages.
- Analytics dashboards for hospital authorities.
- Notification integration for appointment reminders.
- More configurable triage and recommendation rules under clinical governance.
- Cloud deployment and multi-hospital scaling.
- Expanded accessibility features and mobile-first support.

16. Conclusion

- Patient Case-Taking Software using AI provides a structured digital workflow for Mangaluru hospitals.
- It combines patient case capture, AI-assisted questioning, evidence handling and appointment scheduling.
- Doctor capacity and availability are incorporated into assignment and recommendation workflows.
- Patients can select appointment times and interact through multiple Indian/local languages.
- Role-based patient, doctor and hospital-authority access supports a coordinated healthcare workflow.
- The platform is designed as an assistive system, with clinical decisions remaining under qualified healthcare professionals.

THANK YOU

Patient Case-Taking Software using AI
(Mangaluru Hospitals)

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